

CG2028 Assignment - EWMA Open Test Cases

Required assembly function: **int ewma_filter(int new_data, int old_output, int alpha_percent);**

Calculation: $\text{output} = [\text{alpha_percent} \times \text{new_data} + (100 - \text{alpha_percent}) \times \text{old_output}] / 100$. Use signed integer arithmetic and truncate division towards zero. For each test case and each axis, old_output starts at 0 and is replaced by the returned output after every sample.

Output format

Each tuple is {X, Y, Z} after one sequential sample has been processed. There are 16 output tuples per test case.

Open test case 1

alpha_percent = 25 (alpha = 0.25)

Test input

int sensor_data_x[16] = {1000,1001,1002,1003,1004,1005,1006,1007,1008,1009,1010,1011,1012,1013,1014,1015};

int sensor_data_y[16] = {1016,1017,1018,1019,1020,1021,1022,1023,1024,1025,1026,1027,1028,1029,1030,1031};

int sensor_data_z[16] = {1032,1033,1034,1035,1036,1037,1038,1039,1040,1041,1042,1043,1044,1045,1046,1047};

Expected output after each sample

Sample	X	Y	Z
1	250	254	258
2	437	444	451
3	578	587	596
4	684	695	705
5	764	776	787
6	824	837	849
7	869	883	896
8	903	918	931
9	929	944	958
10	949	964	978
11	964	979	994
12	975	991	1006
13	984	1000	1015
14	991	1007	1022
15	996	1012	1028
16	1000	1016	1032

Tuple sequence: {250,254,258} , {437,444,451} , {578,587,596} , {684,695,705} , {764,776,787} , {824,837,849} , {869,883,896} , {903,918,931} , {929,944,958} , {949,964,978} , {964,979,994} , {975,991,1006} , {984,1000,1015} , {991,1007,1022} , {996,1012,1028} , {1000,1016,1032}

Open test case 2

alpha_percent = 50 (alpha = 0.50)

Test input

```
int sensor_data_x[16] = {0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0};
```

```
int sensor_data_y[16] = {1000,1000,1000,1000,1000,1000,1000,1000,1000,1000,1000,1000,1000,1000,1000,1000};
```

```
int sensor_data_z[16] = {1000,-1000,1000,-1000,1000,-1000,1000,-1000,1000,-1000,1000,-1000,1000,-1000,1000,-1000};
```

Expected output after each sample

Sample	X	Y	Z
1	0	500	500
2	0	750	-250
3	0	875	375
4	0	937	-312
5	0	968	344
6	0	984	-328
7	0	992	336
8	0	996	-332
9	0	998	334
10	0	999	-333
11	0	999	333
12	0	999	-333
13	0	999	333
14	0	999	-333
15	0	999	333
16	0	999	-333

Tuple sequence: {0,500,500} , {0,750,-250} , {0,875,375} , {0,937,-312} , {0,968,344} , {0,984,-328} , {0,992,336} , {0,996,-332} , {0,998,334} , {0,999,-333} , {0,999,333} , {0,999,-333} , {0,999,333} , {0,999,-333} , {0,999,333} , {0,999,-333}

Open test case 3

alpha_percent = 10 (alpha = 0.10)

Test input

```
int sensor_data_x[16] = {6030,6000,4389,-4488,6734,5009,-7318,6040,5000,-5643,3888,3488,-3488,3488,4876,5010};
```

```
int sensor_data_y[16] = {-9800,9800,6573,6753,-5004,5895,4321,-6753,9004,3005,-6934,4444,7981,-5144,5867,4178};
```

```
int sensor_data_z[16] = {5455,-6177,8653,7777,6520,-4566,7860,5199,-6233,5444,9822,-3455,6445,7888,-4113,4998};
```

Expected output after each sample

Sample	X	Y	Z
1	603	-980	545
2	1142	98	-127
3	1466	745	751
4	870	1345	1453
5	1456	710	1959
6	1811	1228	1306
7	898	1537	1961
8	1412	708	2284
9	1770	1537	1432
10	1028	1683	1833
11	1314	821	2631
12	1531	1183	2022
13	1029	1862	2464
14	1274	1161	3006
15	1634	1631	2294
16	1971	1885	2564

Tuple sequence: {603,-980,545} , {1142,98,-127} , {1466,745,751} , {870,1345,1453} , {1456,710,1959} , {1811,1228,1306} , {898,1537,1961} , {1412,708,2284} , {1770,1537,1432} , {1028,1683,1833} , {1314,821,2631} , {1531,1183,2022} , {1029,1862,2464} , {1274,1161,3006} , {1634,1631,2294} , {1971,1885,2564}

Open test case 4

alpha_percent = 75 (alpha = 0.75)

Test input

```
int sensor_data_x[16] = {0,0,0,0,4000,4000,4000,4000,-4000,-4000,-4000,-4000,1000,2000,3000,4000};
```

```
int sensor_data_y[16] = {-2000,-1500,-1000,-500,0,500,1000,1500,2000,1500,1000,500,0,-500,-1000,-1500};
```

```
int sensor_data_z[16] = {100,-100,200,-200,400,-400,800,-800,1600,-1600,3200,-3200,6400,-6400,1280,-1280};
```

Expected output after each sample

Sample	X	Y	Z
1	0	-1500	75
2	0	-1500	-56
3	0	-1125	136
4	0	-656	-116
5	3000	-164	271
6	3750	334	-232
7	3937	833	542
8	3984	1333	-464
9	-2004	1833	1084
10	-3501	1583	-929
11	-3875	1145	2167
12	-3968	661	-1858
13	-242	165	4335
14	1439	-333	-3716
15	2609	-833	31
16	3652	-1333	-952

Tuple sequence: {0,-1500,75} , {0,-1500,-56} , {0,-1125,136} , {0,-656,-116} , {3000,-164,271} , {3750,334,-232} , {3937,833,542} , {3984,1333,-464} , {-2004,1833,1084} , {-3501,1583,-929} , {-3875,1145,2167} , {-3968,661,-1858} , {-242,165,4335} , {1439,-333,-3716} , {2609,-833,31} , {3652,-1333,-952}

Open test case 5

alpha_percent = 33 (alpha = 0.33)

Test input

```
int sensor_data_x[16] = {1,2,3,4,5,6,7,8,9,10,-1,-2,-3,-4,-5,-6};
```

```
int sensor_data_y[16] = {99,100,101,102,-99,-100,-101,-102,333,-333,666,-666,999,-999,50,-50};
```

```
int sensor_data_z[16] = {1234,2345,-3456,4567,-5678,6789,-7890,8901,-9012,101,202,-303,404,-505,606,-707};
```

Expected output after each sample

Sample	X	Y	Z
1	0	32	407
2	0	54	1046
3	0	69	-439
4	1	79	1212
5	2	20	-1061
6	3	-19	1529
7	4	-46	-1579
8	5	-64	1879
9	6	67	-1715
10	7	-65	-1115
11	4	176	-680
12	2	-101	-555
13	0	262	-238
14	-1	-154	-326
15	-2	-86	-18
16	-3	-74	-245

Tuple sequence: {0,32,407} , {0,54,1046} , {0,69,-439} , {1,79,1212} , {2,20,-1061} , {3,-19,1529} , {4,-46,-1579} , {5,-64,1879} , {6,67,-1715} , {7,-65,-1115} , {4,176,-680} , {2,-101,-555} , {0,262,-238} , {-1,-154,-326} , {-2,-86,-18} , {-3,-74,-245}